DOHYEON (DANIELLE) KIM

Applied and Computational Mathematics, California Institute of Technology
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RESEARCH INTERESTS

Interacting particle systems and mean-field limits; propagation of chaos, long-time asymptotics, fluctuations around the mean-field limit, and large deviation principles; stochastic differential equations and kinetic equations; mathematical analysis of particle methods for sampling and nonlinear filtering; classical and quantum dynamics for optimization.

EDUCATION

California Institute of Technology, USA *Sep 2021 – Expected Jun 2027*
Ph.D. Candidate in Applied and Computational Mathematics GPA: 4.24/4.3
Advisor: Prof. Franca Hoffmann (since May 2023)

Yonsei University, South Korea
BBA in Business Administration and BS/MS in Mathematics GPA: 4.29/4.3 (4.0/4.0)
Graduated with Highest Distinction; Valedictorian (1/2674)
Master's thesis: "Solvability of Second-Order Elliptic and Parabolic Equations with VMO-Type Coefficients"
Advisor: Prof. Seick Kim

Columbia University, USA
One-year undergraduate study abroad program GPA: 3.9/4.0
Completed eight upper-level mathematics courses

PUBLICATIONS AND PREPRINTS

Authors are listed alphabetically, as is customary in mathematics.

From Consensus-Based Optimization to Particle Swarm Optimization: Convergence Guarantees under Drift-Diffusion Coupling
Franca Hoffmann, **Dohyeon Kim**, Ritvik Teegavarapu
Accepted for poster presentation at the NeurIPS 2026 DynaFront Workshop.
arXiv:2609.17002 (September 2026)

Quantitative Target Convergence and Uniform-in-Time Propagation of Chaos for Langevin-Regularized SVGD
Sayan Banerjee, **Dohyeon Kim**
Submitted, arXiv:2608.28827 (August 2026)

Long-time Stability and Convergence of Particle Swarm Optimization
Giacomo Borghi, Hui Huang, **Dohyeon Kim**
Submitted, arXiv:2607.24696 (July 2026)

Uniform-in-time propagation of chaos for Second-Order Consensus-Based Optimization
Seung-Yeal Ha, Franca Hoffmann, **Dohyeon Kim**
Preprint, arXiv:2605.20697 (May 2026)

Uniform-in-time propagation of chaos for Consensus-Based Optimization
Nicolai Gerber, Franca Hoffmann, **Dohyeon Kim**, Urbain Vaes
Accepted for publication in *SIAM Journal on Mathematical Analysis* (October 2026).
arXiv:2505.08669 (May 2025)

MirrorCBO: A consensus-based optimization method in the spirit of mirror descent
Leon Bungert, Franca Hoffmann, **Doh Yeon Kim**, Tim Roith
Mathematical Models and Methods in Applied Sciences, 2025.
arXiv:2501.12189 (January 2025)

Mono-cluster flocking and uniform-in-time stability of the discrete Motsch–Tadmor model
Seung-Yeal Ha, Franca Hoffmann, **Dohyeon Kim**, Wook Yoon
Communications on Pure and Applied Analysis **32** (2026), 234–281.
doi:10.3934/cpaa.2026040; arXiv:2408.10213 (2024)

MANUSCRIPTS IN PREPARATION

Long-Time Behavior of Continuous-Time Consensus-Based Sampling
Dohyeon Kim
Manuscript in preparation

Refined Long-Time Asymptotics for Variants of Consensus-Based Optimization
Franca Hoffmann, **Dohyeon Kim**, Wuchen Li
Manuscript in preparation.

Mean-Field Well-Posedness and Convergence Guarantees for Localized Consensus-Based Sampling
Arne Bouillon, Kathrin Hellmuth, Franca Hoffmann, **Dohyeon Kim**
Manuscript in preparation.

CONFERENCES, TALKS, AND RESEARCH VISITS

Oberwolfach Workshop: Variational Methods for Evolution *Dec 2026*
Invited participant
Mathematisches Forschungsinstitut Oberwolfach, Germany

Research Visit at BICMR *Oct 17–25, 2026*
Hosted by Prof. Zhenfu Wang
Beijing International Center for Mathematical Research, Peking University, Beijing, China

Interacting Particle Systems and Their Applications *Oct 5–16, 2026*
Invited talk: "Uniform-in-time propagation of chaos and long-time asymptotics for CBO"
Institute for Mathematical Sciences, National University of Singapore, Singapore

Partial Differential Equations for Many Particles Systems *Aug 2026*
Poster: "Uniform-in-time Propagation of Chaos for First- and Second-Order CBO"
ICERM, Brown University, Providence, RI, USA

BIRS–IASM Workshop: Interacting Particle Systems: Theoretical Innovations and Practical Applications *Jun–Jul 2026*
Invited talk (virtual): "Uniform-in-time Propagation of Chaos for First- and Second-Order CBO"
Institute for Advanced Study in Mathematics, Hangzhou, China

13th Summer School on Methods and Models of Kinetic Theory (M&MKT 2026) *Jun 2026*
Poster: "Uniform-in-time Propagation of Chaos for First- and Second-Order CBO"
Pesaro, Italy

SIAM Conference on Uncertainty Quantification (UQ26) *Mar 2026*
Invited talk: "Uniform-in-time Propagation of Chaos for Consensus-Based Optimization"
Minneapolis, MN, USA

Pathways Workshop & Introductory Workshop: Kinetic Theory & Stochastic Partial Differential Equations *Aug 2025*
Participant, Simons Laufer Mathematical Sciences Institute, Berkeley, CA, USA

Frontiers in Applied Analysis *Jun 2025*
Poster: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
Carnegie Mellon University, Pittsburgh, PA, USA

SIAM Conference on Applications of Dynamical Systems *May 2025*
Minisymposium organizer and speaker
Talk: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
Minisymposia MS 47, 65, 83: "Collective Dynamics in Multi-Agent Systems: Advances in Learning and Optimization"
Denver, CO, USA

AIM Workshop: Integro-differential Equations in Many-Particle Interacting Systems *Apr 2025*
Participant, American Institute of Mathematics, Pasadena, CA, USA

SOCAMS 2025 *Apr 2025*
Invited talk: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
University of California, Riverside, CA, USA

SIAM Conference on Mathematics of Data Science *Oct 2024*
Poster: "Recent Developments in Consensus Based Optimization (CBO)"
Minisymposium MS 27: "Interacting Particle Systems in Data Science: From Theory to Applications"
Atlanta, GA, USA

Junior Kinetic Theory Workshop *Sep 2024*
Invited talk: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
Yonsei University, South Korea

9th European Congress of Mathematics (ECM) *Jul 2024*
Invited talk: "Uniform-in-time Mean Field Limit for Consensus-Based Optimization (CBO)"
Seville, Spain

Modern Perspectives in Applied Mathematics: Theory and Numerics of PDEs *Jul 2024*
Participant, ETH Zurich, Switzerland

CIRM: Mathematical Biology: Collective Behavior and Pattern Formation *Jul 2024*
Poster: "Mono-cluster flocking and uniform-in-time stability of the discrete Motsch–Tadmor model"
CIRM, Marseille, France

CIRM: Frontiers in Interacting Particle Systems, Aggregation-Diffusion Equations & Collective Behavior *Jun 2024*
Participant, CIRM, Marseille, France

SOCAMS 2024 *Apr 2024*
Talk: "Recent developments in Consensus-Based Optimization (CBO)"
University of California, San Diego, CA, USA

International Congress on Industrial and Applied Mathematics (ICIAM) *Aug 2023*
Participant, Waseda University, Tokyo, Japan

Purpose-Driven Particle Systems *Mar 2023*
Participant, Lorentz Center, Leiden, Netherlands

HONORS & AWARDS

Graduation with Highest Distinction
Graduation ceremony valedictorian — Yonsei University, South Korea

Amore Pacific President Award
Alumni Association Presidential Award — Yonsei University, South Korea

Silver Prize
National Math Olympiad for Undergraduate Students, Korean Mathematical Society, South Korea
Non-mathematics-major division for engineering, medical, and other science students

Awards of Highest Academic Honor and Academic Honor
Top 1% and top 3% of students in the department, respectively
Yonsei University, South Korea

FELLOWSHIPS & GRANTS

Oberwolfach Travel Grant *Dec 2026*
Support for the workshop "Variational Methods for Evolution"
Mathematisches Forschungsinstitut Oberwolfach, Germany

BICMR Research Visit Lodging Support *Oct 2026*
Host-provided lodging support for a research visit with Prof. Zhenfu Wang
Beijing International Center for Mathematical Research, Peking University, China

IMS Lodging Support *Oct 2026*
Support for "Interacting Particle Systems and Their Applications"
Institute for Mathematical Sciences, National University of Singapore, Singapore

ICERM Lodging Support *Aug 2026*
Support for "Partial Differential Equations for Many Particles Systems"
ICERM, Brown University, USA

M&MKT Summer School Lodging Support *Jun 2026*
Support for the 13th Summer School on Methods and Models of Kinetic Theory
M&MKT, Pesaro, Italy

Caltech Y Hummel Gray Award *Nov 2025*
Travel award for graduate students — Caltech, USA

Travel Support for SLMath Workshops *Aug 2025*
Support to attend the Pathways Workshop and Introductory Workshop on Kinetic Theory & Stochastic Partial Differential Equations
Simons Laufer Mathematical Sciences Institute, Berkeley, CA, USA (workshop venue)

CMU Travel Support *Jun 2025*
Support for "Frontiers in Applied Analysis" — Carnegie Mellon University, USA

SIAM Travel Award *May 2025*
Support for the Conference on Applications of Dynamical Systems — SIAM, USA

ETH Zurich Conference Travel Support *Jul 2024*
Support for "Modern Perspectives in Applied Mathematics: Theory and Numerics of PDEs"
ETH Zurich, Switzerland

CIRM Conference Travel Support *Jun–Jul 2024*
Travel and lodging support for "Frontiers in Interacting Particle Systems" and "Mathematical Biology: Collective Behavior and Pattern Formation"
CIRM, Marseille, France

Two-Year Departmental Fellowship *Sep 2021 – Sep 2023*
Computing and Mathematical Sciences departmental fellowship — Caltech, USA

OSE Tutoring Scholarship
Award for successful tutoring — OSE Center, Yonsei University, South Korea

Merit-Based Undergraduate/Graduate Scholarship
External scholarship — Kwanjeong Foundation, South Korea

Merit-Based Undergraduate Scholarship
External scholarship
Korean Foundation for Advanced Studies (KFAS), SK, South Korea

Academic Scholarship
University and private-fund scholarships — Yonsei University, South Korea

Academic Scholarship (declined)
Korea University Business Department, South Korea
Awarded to admitted undergraduates with exceptional academic achievements

TEACHING EXPERIENCE

All teaching assistant appointments below were at the California Institute of Technology, USA.

TA, ACM 270: Particles in Motion—Mathematical Modelling Across Scales *Apr–Jun 2026*
Instructor: Prof. Kathrin Hellmuth; graduate-level course
Developed lecture notes and delivered two 1.5-hour lectures on mean-field limits and propagation of chaos; held weekly office hours; developed and graded homework problems.

TA, ACM 95/100A: Introductory Methods of Applied Mathematics for the Physical Sciences *Jan–Mar 2026*
Instructor: Prof. Konstantin Zuev; undergraduate/graduate-level course
Held weekly office hours; graded weekly homework problems and exams.

TA, ACM 256: Optimal Transport *Jan–Mar 2025*
Instructors: Prof. Franca Hoffmann and Prof. Ricardo Baptista; graduate-level course
Held weekly office hours; graded homework problems.

TA, ACM 95/100A: Introductory Methods of Applied Mathematics for the Physical Sciences *Jan–Mar 2024*
Instructor: Prof. Franca Hoffmann; undergraduate/graduate-level course
Held weekly office hours; graded weekly homework problems and exams.

TA, CMS/ACM 117: Probability Theory and Stochastic Processes *Sep–Dec 2023*
Instructor: Prof. Joel Tropp; graduate-level course
Graded weekly homework problems and exams.

TA, ACM 105: Functional Analysis *Jan–Mar 2023*
Instructor: Prof. Oscar Leong; graduate-level course
Delivered two 1.5-hour lectures on the Hahn–Banach theorem and the open mapping and closed graph theorems; held weekly office hours; graded homework problems and exams.

TA, CMS/ACM 117: Probability Theory and Stochastic Processes *Sep–Dec 2022*
Instructor: Prof. Joel Tropp; graduate-level course
Held weekly office hours; wrote problem-set solutions; graded weekly homework problems and exams.

Private Tutoring
Tutored single- and multivariable calculus for a first-year undergraduate — Seoul, South Korea

Peer Tutoring
Tutored undergraduate peers in Analysis 1 — OSE Center, Yonsei University, South Korea

MENTORSHIP

Ritvik Teegavarapu, Caltech SURF mentee, USA *May 2024 – present*
Co-mentored with Prof. Franca Hoffmann on particle swarm optimization; continuing collaboration on annealed consensus-based optimization and sampling
Student awards: SOCAMS Best Poster Award (2025); Caltech's George W. and Bernice E. Green Memorial Prize (research award, 2026)
Now a Ph.D. student with Prof. Kavita Ramanan at Brown University (since fall 2026)

LEADERSHIP & ACADEMIC SERVICE

SIAM Student Chapter, Caltech, USA *2025 – present*
President (2025–26); committee member (2026–27)

Womxn in Engineering and Applied Science, Caltech, USA *2024 – present*
Committee member; organized kick-off events, faculty lunches and coffee conversations, and social gatherings.

Research Group Meeting Organizer, Caltech, USA *Spring 2025*
Organized meetings for Prof. Franca Hoffmann's research group.

Head Proctor, First-Year Preliminary Exam, Caltech, USA *Winter 2023*
Coordinated mock exams, matched students with proctors, and collected and shared past exams.

OUTREACH & SCIENCE COMMUNICATION

Three Minute Thesis (3MT) Competition, Caltech, USA *April 2026*
Finalist; the only finalist from Mathematics or Applied and Computational Mathematics in 2026.
Presented consensus-based optimization research to a general audience.

AIM Math Fair, American Institute of Mathematics, USA *June 2025*
Volunteer; guided visiting families, from young children to adults, through mathematical puzzles.

PROGRAMMING & LANGUAGES

Programming and typesetting: Python, MATLAB, LaTeX
Languages: Korean (native); English (fluent); French (intermediate; DELF B1, B2)